INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

OVEN, BAKING AND ROASTING, FORCED CONVECTION, ELECTRIC (NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

This commercial item description (CID) covers single-compartment and two-compartment, baking and roasting, electric, forced convection ovens. These ovens are intended for use in Naval shipboard galleys and bakeries where combined use and time savings are required.

2. CLASSIFICATION

Type 1: Single compartment
Type 2: Two compartments

Size 1: Standard depth oven cavity
Size 2: Extra deep oven cavity

3. SALIENT CHARACTERISTICS

3.1 Design and construction. The front, sides, back, and top exterior surfaces of the type 1 oven shall be stainless steel; the bottom surface shall be carbon steel or stainless steel. The front, sides, and back exterior surfaces of the type 2 oven shall be stainless steel; the top and bottom surfaces shall be carbon steel or stainless steel.

The type 1 oven shall provide the following features:

- Single oven body
- Removable stainless steel oven compartment linings
- Vertically split doors (side opening type) with handles
- Five equally spaced shelves capable of supporting a load of 20 pounds per square foot
- Removable shelves and shelf supports

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, 2531 Jefferson-Davis Highway, Arlington, VA 22242-5160.

AMSC N/A FSC 7310
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- Two-speed oven blower to provide circulation of forced, heated air throughout the oven compartment (which becomes de-energized when the door is opened)
- Blower switch (low and high speeds)
- Adjustable thermostat
- 5-hour timer with alarm, installed in control panel or oven front
- Oven cavity light
- Indicator light (power)
- ON-OFF power switch
- Adjustable air baffle
- Remotely operated vent to control and release moisture from the baking chamber (operated from control panel)

The type 2 oven shall consist of two identical single ovens (same as type 1), stacked and secured together.

The oven shall preheat to 350°F (starting from ambient air temperature of approximately 70°F) in 10 minutes. Each oven compartment shall operate independently with a separate thermostat. Both type 1 and type 2 ovens shall support a uniformly distributed load of 500 pounds without any visible permanent deformation. A terminal board or box shall be provided to make permanent connections to the installation wiring system, and it shall be grounded to the oven shell. Controls, switches, and indicator lights shall be located in a control panel at the front of the oven.

3.2 Heating units. The oven shall be provided with heating elements located to provide even heat distribution throughout the oven compartment. The heating elements shall be fully protected and removable from the oven front. The heating elements shall be energized by heat control switches.

3.3 Dimensions. Oven dimensions shall be within the limits given in Table 1.

TABLE 1. Dimensions (inches).

Measurements	Size 1		Size 2	
	Minimum	Maximum	Minimum	Maximum
Width, overall	33	40	33	40
Depth, overall	32	39	38	46
Height, overall (excluding vent) (1)				
Type 1	54	65	54	69
Type 2	57	78	52	78
Height, from bottom of lower oven to deck				
Type 1	26	38	26	38
Type 2	11	17	11	17
Compartment (inside) (2)				
Width	26	30	26	30
Depth	19	25	27	38
Height	19	22	19	22

Notes:
1. Height from finished deck
2. Each shelf/rack of size 1 ovens shall accommodate a nominal 18 x 26-inch sheet pan or roasting pan. Each shelf/rack of size 2 ovens shall accommodate two five-strap bread pans. Pans specified shall not make contact with oven interior surfaces, and contents of the load pans shall not contact oven surfaces, shelves, or racks during roasting or baking operations.

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3.4 Removable oven lining panels. Removable oven lining panels shall be stainless steel. Panels shall be made removable without the use of tools, if possible, but the use of basic tools (e.g., screwdriver, pliers, adjustable wrench) is acceptable.

3.5 Oven door latch. Oven cavity door interiors shall be stainless steel. Oven doors shall be equipped with secondary exterior mechanical latches (in addition to the main interior door latches). Each side-opening-type door shall be provided with a hold-open latch which holds the doors in a wide-open position flush with the oven interior side walls.

3.6 Mounting. The type 1 oven shall be provided with an open stainless steel stand suitable for secure attachment to the deck. The type 2 oven shall be provided with four stainless steel legs not less than 8 inches high suitable for secure attachment to the deck. (Shipboard requirement, see 7.2)

3.7 Inclined operation and structural integrity. The unit shall operate satisfactorily, in accordance with the requirements of this CID (such as the door remaining closed to prevent the loss of contents), when test-operated for 30 seconds inclined at an angle of 15° each side of the vertical, in each of two vertical planes at right angles to each other. The oven shall not deviate or rack from perpendicularity to the deck by more than 1/8 inch when the deck is inclined at an angle of 15°. (Shipboard requirement, see 7.2)

3.8 Environmental suitability. The unit shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts, and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage, when tested in accordance with MIL-STD-167-1, type I equipment. The unit shall be secured to the test machine in the same manner that it will be secured on shipboard. (Shipboard requirement, see 7.2)

3.9 Electrical requirements. The unit shall operate on 440 VAC, 60 Hz, three-phase power, as defined in MIL-STD-1399, Section 300, and shall have provisions for making direct (hardwired) connections for electric power (2 conductors) and for equipment grounding (1 conductor). All outermost metallic surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be not greater than 0.1 ohm. Electrical components, other than the hermetically sealed motor, shall be provided in accordance with NEMA 250, Type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45. (Shipboard requirement, see 7.2)

3.10 Heating elements. The ends of the heating elements shall be sealed and protected to preclude the entrance of moisture. Heating elements shall be removable from within the oven cavity. Leads requiring disconnection from terminal boards or strips outside the oven shall not be used.

3.11 Accessibility. The unit shall be physically and visually accessible from the front for operation and for normal maintenance with tools, test equipment, and replacement parts. No special tools shall be required for access or parts replacement.

3.12 Oven cavity light. The oven interior shall be provided with a light which is recessed and protected with a heat-resistant window. The light shall be activated by either a manual switch or an automatic door switch.

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3.13 Physical size limitations. The unit shall pass or shall disassemble so that it can pass through a shipboard access measuring 26 x 66 inches with 8-inch radius corners. (Shipboard requirement, see 7.2)

3.14 Label plates. The unit shall be provided with a data nameplate and an instruction plate, both attached to the front of the unit. They shall be readily visible during normal operating use and shall not adversely affect the life and utility of the unit.

3.14.1 Data nameplate. The data nameplate shall contain the manufacturer's name, model, serial number, date manufactured, and any other information needed to uniquely identify the unit.

3.14.2 Instruction plate. The instruction plate shall provide instructions for start-up, operation, and shut down.

3.15 High voltage labels. A label reading "Danger—High Voltage" shall be affixed to the outer case assembly, on or adjacent to each service access cover near one of the fasteners securing the cover. In addition, a warning label in accordance with UL 969 shall be placed near the high voltage components inside the equipment. This label shall include, but not be limited to the following texts:

- Danger—High Voltage.
- Power supply must be disconnected before servicing.
- Access covers must be in place before use.
- Service should be performed by authorized personnel only.

3.16 Standards compliance. The unit shall be in accordance with and certified to meet the applicable requirements of NSF 4 and UL 197.

4. REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market, or the same product that has been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such compliance.

6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Ordering data.

- Title, number, and date of this CID
- Type and size required
- When required, manuals shall be in accordance with ASTM F 760

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7.2 Shipboard requirement. Whenever a "(Shipboard requirement)" is included in a paragraph under SALIENT CHARACTERISTICS, it is meant that the requirement is something that is not normally offered to the commercial market by the manufacturer.

7.3 Source of documents.

7.3.1 Military documents. Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally excited)
MIL-STD-1399 - Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating

7.3.2 American Society for Testing and Materials (ASTM) Standards. ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959.

ASTM F 760 - Food Service Equipment Manuals

7.3.3 National Sanitation Foundation International (NSF) Standards. NSF Standards are available from the National Sanitation Foundation International, 3475 Plymouth Road, P.O. Box 130140, Ann Arbor, MI 48113-0140.

NSF 4 - Commercial Cooking, Rethermalization and Powered Hot Food Holding and Transport Equipment

7.3.4 Underwriters Laboratories (UL) Standards. UL Standards are available from the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.

UL 197 - Commercial Electric Cooking Appliances
UL 969 - Marking and Labeling Systems

7.3.5 The Institute of Electrical and Electronic Engineers (IEEE) Standards. IEEE Standards are available from The Institute of Electrical and Electronic Engineers, 445 Hose Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard

7.4 Suggested sources of supply. Manufacturers of products known to meet the requirements of this CID are listed below. However, competition is not limited to these companies.

G. S. Blodgett Company
50 Lakeside Av.
Burlington, VT 05401

Lang Manufacturing Co.
6500 Merrill Creek Parkway
Everett, WA 98201

Vulcan-Hart Corp.
3600 North Point Blvd.
Baltimore, MD 21222-2726

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MILITARY INTERESTS:

Custodian:
Navy - SH
Army - GL

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing activity:
Navy - SH
(Project 7310-0891)

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